

Random Variable

- Summary ✓
- Agenda ✓
- Next ←

x, y, z } possible value of this variable
are numerical

Discrete

Continuous

$[1, 10]$

Runs

$[0, 1]$

Strike rate

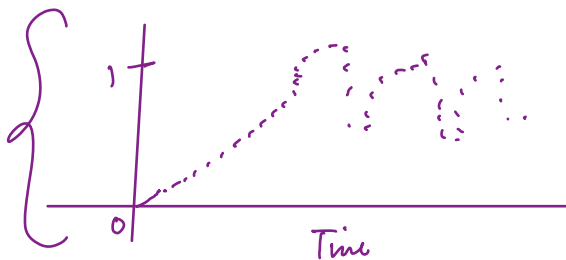
Count of matches, where Sachin scored a century.

↳ Discrete

What's Sachin's strike rate in a match?

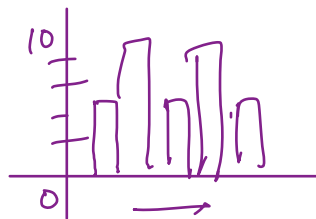
↳ Continuous.

$[0, 1]$



$z_0 \rightarrow z_1$

$[No. of centuries]$



Probability function

① Probability Mass function

↳ discrete random variable

Eg ∇ Probability that Sachin scores exactly 1 century in a match

DRV

1 century

Eg ∇ Sachin hits exactly 4 boundaries in a match

DRV

4 boundaries

Fair Die $\rightarrow [1, 2, 3, 4, 5, 6] \rightarrow 1/6$

$$P(\underset{\substack{\downarrow \\ \text{DRV}}}{X} = x) = p(x)$$

Binomial Distribution

Models the no. of successes (K) in n independent trials with p (success probability)

$$\begin{aligned} n &= 10 \rightarrow \left. \begin{aligned} \frac{1}{2} H &= 5 \\ \frac{1}{2} T &= 5 \end{aligned} \right\} \end{aligned}$$

$$\begin{aligned} &7 \rightarrow H \quad 3 \rightarrow T \\ &p = 0.3 \end{aligned}$$

$$P(\underline{X} = \underline{k}) = \binom{\overset{n=10}{\underline{n}}}{\underline{k}} p^k (1-p)^{n-k}$$

$$\left\{ \binom{n}{k} = \frac{n!}{k! (n-k)!} \right.$$

$\downarrow$
7
 $\downarrow$
0.3

Model the no. of events occurring in a
fixed interval

(email per hour)
 + Cars passing through a toll booth in
 a day
 + Defects in a metre of fabric

+ Poisson Distribution

$$P(\underline{X} = \underline{k}) = \frac{1^{\lambda} e^{-1}}{k!}$$

$\lambda \rightarrow$ Average rate of events (5 calls/day)

$K \rightarrow$ no. of occurrences (7 calls)

$L \rightarrow 2.71$

Q Suppose a hospital receives 4 emergency calls per hour
 $\lambda =$
Prob of receiving exactly 2 calls in the next hour.

Ans

$$P(X=2) = \frac{4^2 e^{-4}}{2!} \Rightarrow 0.148$$

$\Rightarrow 0.15$

where to use poisson

- ① Rare event $\rightarrow$ accidents / system failures
- ② fixed intervals

n is large
 p is small } $\Rightarrow$ binomial distribution

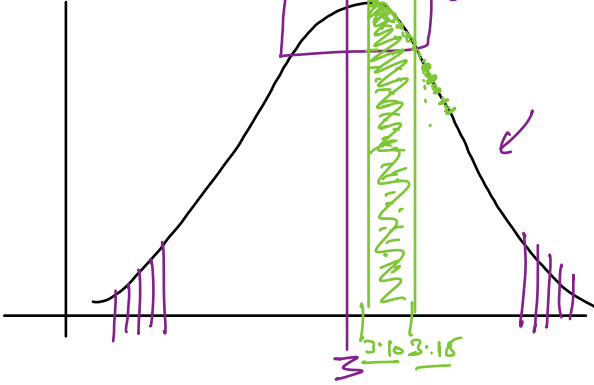
Continuous Random Variables

How crowded the chances are

↳ Probability Density function

PDF

↳ it does not give me probability directly



0.73

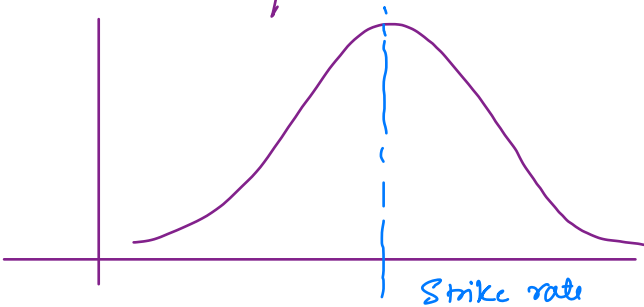
Ice Cream Truck

→ 3 pm

eg Model the distribution of Sachin's strike rate

eg Model the dist of Sachin's time on field.
Continuous

Normal distribution / bell curve



$$pdf = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$\sigma \rightarrow \text{std}$

$\mu \rightarrow \text{mean}$

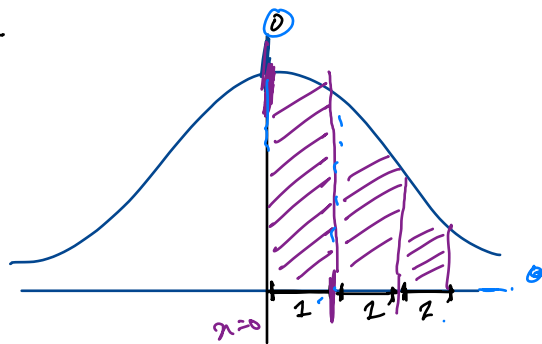
$e = 2.71$

$\text{CIT} \rightarrow \boxed{Z \rightarrow \underline{Z\text{-score}}} \Rightarrow \frac{X-\mu}{\sigma} \rightarrow \text{Deviation from mean}$

PDF $\rightarrow$ Standard normal dist

$\rightarrow \underline{\mu=0}$

$\rightarrow \underline{\sigma=1}$



PDF = $\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$

$$\begin{aligned} &0.39 \\ &+ 0.24 \\ &+ 0.05 \\ &\hline &0.68 \end{aligned}$$

$x=0$

$$f(0) = \frac{1}{\sqrt{2\pi}} e^{-0}$$

$$\Rightarrow \frac{1}{\sqrt{2\pi}}$$

$$\approx 0.39$$

pdf has decreased

$x=1$

$$f(1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1^2}{2}}$$

$$\Rightarrow \frac{1}{\sqrt{2\pi}} e^{-1/2}$$

$$\approx 0.23$$

pdf has decreased

$$x=2 \quad f(2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}} = \frac{1}{\sqrt{2\pi}} e^{-2} = \underline{\underline{0.05}}$$

$$\underline{\underline{x=2}} \quad \checkmark$$

Cumulative Distribution function

→ discrete & continuous Random Variables

$$P(X \leq \underline{\underline{x}})$$

→ The prob that Sachin scores upto 2 ^{DRV} centuries in a match

→ The prob that Sachin's SK ≤ 80 _{CRV} in a match

Bernoulli Dist

↳ 2 outcomes → Success / failure
→ single trial

Success → Sachin scores a century
failure - Sachin doesn't score a century